Predicting Education Fee Burden and Affordability Stress Among Indian Households Using NSS and CMS Education Microdata

Synopsis

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In India, education is widely seen as a key path for social mobility and long-term improvement in living standards, especially for families from lower and middle income groups. At the same time, the total cost of schooling has become a major component of household budgets, including tuition fees as well as books, uniforms, transport, private coaching, hostel or mess charges, and other related expenses. When education expenses become too high compared to overall household consumption expenditure, families may be forced to cut back on essential items, take informal loans, shift children to cheaper schools, or even discontinue education in extreme cases. The resulting financial and emotional stress can affect both education continuity and overall household welfare.

The National Sample Survey (NSS) 75th Round on household social consumption: education and the more recent Comprehensive Modular Survey – Education (CMS-E 2025) under the 80th Round provide nationally representative microdata on household characteristics, education status, and detailed education-related expenditures. These official datasets are designed for public use and are widely used in policy analysis and academic work. This capstone project proposes to move beyond simple descriptive tables and develop an explainable predictive model at the household level for education fee burden and affordability stress using these data sources.

Background and Context

In recent years, official and independent reports have highlighted that education spending in India is rising faster than many other components of household consumption, especially in urban areas and among families that opt for private schooling and coaching. Parents often bear not only regular school fees but also a wide range of additional costs, such as books, uniforms, transport, private tuition, exam fees, hostel charges, and miscellaneous contributions. When these costs consume a large share of monthly resources, households may face affordability stress and difficult trade-offs between education and other essential expenditures such as food, health, or housing.

The burden of education costs is not evenly distributed. Education choices and expenses vary substantially across states, between rural and urban areas, across social groups, and across different types of institutions (government, aided, private unaided, and low-fee private schools). There are also differences by level of education, medium of instruction, and use of private coaching, which is increasingly common in many parts of India. Policy makers and practitioners need more granular evidence on which types of households are most at risk of high education fee burden and which specific factors drive this burden, so that schemes like scholarships, fee waivers, subsidies, or information campaigns can be better targeted.

The NSS 75th Round (Schedule 25.2) on household social consumption: education and the CMS-E 2025 survey under the 80th Round of NSS provide detailed expenditure and consumption information for large, nationally representative samples of Indian households. These datasets include household-level consumption expenditure, school attendance, institution type, and education-related expenses. This combination makes it possible to construct a quantitative indicator of household-level education fee burden and to model the risk of high burden using both statistical and machine learning approaches.

Problem Statement

Most existing official reports and academic studies on education expenditure in India have focused on descriptive statistics, such as average spending by state, social group, or type of institution, and on explaining determinants of education expenditure at an aggregate level. There is relatively less work that directly predicts, at the household level, whether a given family is likely to face high education fee burden based on its characteristics and education choices, and that presents the results in an explainable way suitable for policy and programme use.

This capstone project addresses the following practical problem:

How can official NSS and CMS education microdata be used to predict whether a household faces high education fee burden, defined as education spending exceeding a chosen threshold share of household consumption expenditure, and which household and education-related characteristics are most strongly associated with this burden?

Purpose of the Study

The main purpose of this study is to develop and evaluate an explainable predictive model for education fee burden and affordability stress among Indian households using official survey microdata from NSS 75th Round and CMS-E 2025. More specifically, the study aims to:

1. Construct a household-level indicator of education fee burden, defined as the ratio of total education-related expenditure to usual household consumption expenditure.
2. Classify households into “high fee burden” and “non-high fee burden” categories based on a policy-relevant threshold (for example, 10 percent of usual household consumption expenditure).

3. Use statistical and machine learning techniques to model the probability that a household will fall into the high fee burden category, using both household characteristics and education-related factors as predictors.

4. Provide interpretable insights into which variables are most important in explaining and predicting high education fee burden, so that the results are useful for real-world decision-making rather than remaining a black-box exercise.

Scope and Objectives

This project will focus on Indian households covered in the NSS 75th Round education survey (July 2017–June 2018) and the CMS-E 2025 education dataset (April–June 2025). The analysis will be restricted to households in which at least one member is attending an educational institution and for which key education expenditure and household consumption variables are available.

The specific objectives of this study are:

1. To construct a robust measure of education fee burden at the household level using official NSS and CMS education microdata.

2. To build and compare predictive models that classify households into high and non-high education fee burden categories.

3. To identify the main household and education-related characteristics that are associated with high education fee burden.

4. To generate explainable results (for example, through feature importance and SHAP values) that can guide policy and programme design.

This scope is practical for an 11-week capstone because the data are already collected and documented, the variables are measurable from the survey instruments, and the core tasks (data preparation, modelling, interpretation, and reporting) fit within the course timeline.

Research Questions

The study will address the following research questions:

RQ1. What proportion of households in the NSS and CMS education datasets experience high education fee burden when burden is defined as education expenditure exceeding a given threshold share of household consumption expenditure?

RQ2. How well can household-level characteristics (such as household size, household type, religion, social group, rural or urban sector, and usual consumption expenditure) predict whether a household will experience high education fee burden?

RQ3. How do education-related characteristics (such as institution type, course level, medium of instruction, use of private coaching, and distance to school) influence the probability of a household having high education fee burden?

RQ4. Which predictors are most important in explaining high education fee burden, and how can explainable machine learning methods help communicate these drivers to stakeholders?

Hypotheses

For the main classification problem (high fee burden versus non-high fee burden), the working hypotheses are:

Null hypothesis (H0). There is no statistically significant relationship between the selected household and education characteristics (for example, household size, social group, rural

or urban sector, institution type, use of private coaching, usual household consumption expenditure) and whether a household experiences high education fee burden.

Alternative hypothesis (H1). There is a statistically significant relationship between the selected household and education characteristics and whether a household experiences high education fee burden; these variables meaningfully improve the ability to explain and predict high fee burden.

Sample Size Calculation

The proposed study will use secondary microdata from the NSS 75th Round and the CMS-E 2025 education survey, both of which already contain much larger samples than the minimum required for this capstone. The aim of the sample size calculation is therefore to demonstrate that the available data are adequate for answering Research Questions 1–4 at conventional levels of confidence and power.

Unless otherwise stated, all calculations assume a two-sided significance level of 0.05 and desired statistical power of 0.80, with a medium effect size, which is standard in applied social statistics.

RQ1: Proportion of high-burden households

For RQ1, the objective is to estimate the proportion of households experiencing high education-fee burden with a 95 percent confidence interval and a margin of error of approximately 3 percentage points. Using the usual formula for a single proportion,

$$\boldsymbol{n} = \frac{(\boldsymbol{Z^2} \times \boldsymbol{p} \times (\boldsymbol{1} - \boldsymbol{p}))}{\boldsymbol{e^2}}$$

where Z is the 1.96 critical value for $\alpha = 0.05$, p is a planning value for the true proportion, and e is the desired margin of error. Taking a conservative value $p = 0.50$ and $e = 0.03$, the minimum required sample size is about 1,067 households.

RQ2 and RQ3: Logistic regression and group comparisons

For RQ2 and RQ3, the main analyses will use binary logistic regression models and related chi-square tests of association between high fee burden and key categorical predictors such as social group and institution type. Power calculations for logistic regression with a binary outcome and a medium effect size (for example, an odds ratio of around 1.5 for key predictors) indicate that a total sample of roughly 800 to 1,000 households is adequate to achieve 80 percent power at $\alpha = 0.05$ for the planned model complexity.

RQ4: Events-per-variable rule for predictor importance

For RQ4, the focus is on obtaining stable estimates of predictor importance in logistic and tree-based models. A widely used rule of thumb recommends at least 10 to 20 'events' (households with high fee burden) per predictor variable in logistic regression. Given the anticipated prevalence of high-burden households, this guideline again points to a minimum total sample size of the order of 1,000 or more households, depending on the final number of predictors included.

Final required sample size and feasibility

To be conservative, the final minimum required sample size for this study is taken as the maximum of the above requirements, namely approximately 1,067 households. In practice, the analytic sample will be determined by the number of eligible households in the merged NSS and CMS datasets (households with at least one member currently attending an educational institution and with complete information on education expenditure and total consumption). The

realised sample size in these official surveys is expected to be several times larger than this minimum, so the available data should provide adequate statistical power for all four research questions within the 10-week capstone timeline.

Data Description

The study will use two main sources of public microdata.

1. NSS 75th Round (2017–2018), Schedule 25.2: Household Social Consumption – Education

This survey covers household social consumption on education for the period July 2017 to June 2018 and is conducted by the National Sample Survey Office, Ministry of Statistics and Programme Implementation. Key components are:

- Block 3: Household characteristics – contains household-level variables such as household size, household type, religion, social group, distances to different levels of school, availability of a computer in the household, internet access for any member, whether any household member aged 3–35 is attending an educational institution, and usual monthly household consumption expenditure.
- Block 5: Education particulars for basic course (age 3–35, currently attending) – provides information on individuals' education details, including age, course and level, type of course, institution type, nature of institution (for example, government, aided, private), reasons for opting for private institutions, and scholarship or fee-waiver information.
- Block 6: Particulars of education expenditure for basic course – gives a break-up of education-related expenses for the same individuals, namely course fees, books, stationery and

uniforms, transport, private coaching, and other education expenditure, along with information on funding sources.

These blocks will be merged at the household and person level using common identifiers. Individual education expenditures will then be aggregated to the household level to compute total education spending for each household.

2. CMS-E 2025 Household Education Survey (80th Round, CMS: Education)

The CMS-E 2025 survey is part of the 80th Round and provides updated information on household education expenditure and attendance for April–June 2025. The CMS household file contains variables such as survey year, sector (rural or urban), NSS region, district, household size, household type, religion, social group, whether any member is in hostel or mess, whether any member is attending school, number of members attending school, detailed consumption expenditure categories, and usual monthly household consumption expenditure.

Using these two datasets, a household-level education fee burden ratio will be constructed as:

$$\boldsymbol{education\ fee\ burden} = \frac{\boldsymbol{total\ annual\ education\ expenditure}}{\boldsymbol{usual\ annual\ (or\ annualised)\ household\ consumption\ expenditure}}$$

Households whose ratio exceeds a chosen threshold (for example, 0.10) will be labelled as “high fee burden” for the main classification task.

Data Dictionary

This study uses two primary microdata sources: (i) the CMS Education NSS 80th Round household file (CMSE80HH25.csv) for recent household-level consumption and schooling

characteristics, and (ii) NSS 75th Round Household Social Consumption: Education microdata for detailed education expenditure by student and household.

Key analysis variables

The table below summarises the main variables that will be used in the analysis dataset (after cleaning and integration), including their names, labels, types, and units or allowed values.

Analysis variable name	Label / description	Type	Units / allowed values	Source
household_id	Unique household identifier, constructed from survey ID, region and sample household number so that each row corresponds to one household in	Categorical (string)	Concatenation of survey and household fields (e.g., surveyyear + nssregion + hhldno)	CMS 80th Round, NSS 75th RoundCMSE80HH25.csv+1
survey_year	Survey reference year for the household	Numeric (integer)	Year (e.g., 2017â€“2018 for NSS 75th, 2024â€“2025 for CMS 80th)	CMS and NSS survey year variablesCMSE80HH25.csv+1
sector	Rural or urban location of the household	Categorical	{Rural, Urban} (coded as 1/2 or R/U in original files)	CMS variable sector; NSS sector variable in Block 3CMSE80HH25.csv+1
nss_region	NSS region code used for sampling stratification	Categorical	Region codes as defined in the sample design	CMS variable nssregion; NSS region or stratum codesCMSE80HH25.csv+1
district	District code where the household is located	Categorical	District codes as per the sample frame	CMS variable district; corresponding NSS district codeCMSE80HH25.csv+1
household_size	Number of members in the household	Numeric (integer)	Count of individuals usually residing in the household	CMS variable householdsize; NSS household size in Block 3CMSE80HH25.csv+1
household_type	Type of household by usual source of income or occupation (e.g., selfâ€'employed, regular wage,	Categorical	Categories as defined in NSS and CMS codebooks	CMS variable householdtype; NSS household type in Block 3CMSE80HH25.csv+1
religion	Religion of the household head	Categorical	Category codes (e.g., Hindu, Muslim, Christian, Sikh, others)	CMS variable religion; NSS religion codeCMSE80HH25.csv+1
social_group	Social group / caste category of the household	Categorical	Category codes (e.g., SC, ST, OBC, Others)	CMS variable socialgroup; NSS socialâ€'group codeCMSE80HH25.csv+1
in_hostel_or_mess	Indicator that at least one household member resides in a hostel / mess	Binary (0/1)	1 if any member in hostel/mess, 0 otherwise	CMS variable inhostelormess (or equivalent)CMSE80HH25.csv
any_member_attending_school	Indicator that at least one household member is currently attending educational institution	Binary (0/1)	1 if yes, 0 if no	CMS variable anymemberattendingschool; NSS Block 5 education attendance indicatorCMSE80HH25.csv+1
num_members_attending_school	Number of household members currently attending educational institution	Numeric (integer)	Nonâ€'negative integer	CMS variable nummembersattendingschool; derived counts from NSS Block 5 personâ€'level recordsCMSE80HH25.csv+1
usual_monthly_consumption_expenditure	Usual monthly household consumption expenditure on all items	Numeric (continuous)	Indian Rupees per month	CMS variable usualmonthlyconsumptionexpenditure; NSS usual monthly consumption expenditure variable in Block 3CMSE80HH25.csv+1
annual_household_consumption	Annual household consumption expenditure derived from usual monthly consumption	Numeric (continuous)	Indian Rupees per year	Derived as 12 Ã— usual_monthly_consumption_expenditure
course_fee_amount	Education expenditure on course/tuition fees for a student over the reference period	Numeric (continuous)	Indian Rupees	NSS Block 5 education expenditure component for course fees (personâ€'level)R75252L02-Block-3-household-characteristics.csv
books_stationery_amount	Expenditure on books, stationery, and related learning material	Numeric (continuous)	Indian Rupees	NSS Block 5 education expenditure component for books and stationeryR75252L02-Block-3-household-characteristics.csv
transport_amount	Expenditure on transport related to attending educational institution	Numeric (continuous)	Indian Rupees	NSS Block 5 component for transport expenditureR75252L02-Block-3-household-characteristics.csv
private_coaching_amount	Expenditure on private tuition, coaching or remedial classes	Numeric (continuous)	Indian Rupees	NSS Block 5 component for private coaching / tuitionR75252L02-Block-3-household-characteristics.csv
other_education_expenditure	Other educationâ€'related expenditure (e.g., uniform, examination fees, miscellaneous)	Numeric (continuous)	Indian Rupees	NSS Block 5 â€œotherâ€ education expenditure components (summed)R75252L02-Block-3-household-characteristics.csv
total_education_expenditure_student	Total education expenditure per student over the reference period	Numeric (continuous)	Indian Rupees	Sum of all relevant studentâ€'level components (course_fee_amount + books_stationery_amount + transport_amount + private_coaching_amount + other_education_expenditure)
total_education_expenditure_hh	Total education expenditure per household, aggregated over all students	Numeric (continuous)	Indian Rupees per year	Householdâ€'level sum of total_education_expenditure_student across all student members in NSS Block 5; used as numerator of burden ratio
fee_burden_ratio	Education fee burden ratio for the household	Numeric	Ratio in [0, 1+] (no unit)	Defined as total_education_expenditure_hh / annual_household_consumption
high_burden	Indicator that the household faces high education fee burden	Binary (0/1)	1 if fee_burden_ratio exceeds a defined threshold (e.g., 0.10), 0 otherwise	Derived binary variable; threshold justified in the methodology section
school_type	Management type of the institution attended (e.g., government, governmentâ€'aided, private	Categorical	Categories as per NSS schoolâ€'type codes	NSS Block 5 variable on type of educational institutionR75252L02-Block-3-household-characteristics.csv
education_level	Current level/class of education for each student (e.g., primary, upper primary, secondary, higher	Categorical / ordered	Ordered categories from primary to tertiary	NSS Block 5 education level variableR75252L02-Block-3-household-characteristics.csv
private_institution	Indicator if the student is enrolled in a private institution	Binary (0/1)	1 for private unaided/aided, 0 for government	Derived from school_type
internet_access	Indicator that the household has internet access	Binary (0/1)	1 if household has internet access, 0 otherwise	NSS Block 3 ICT access variables where available; CMS ICT fields if presentR75252L02-Block-3-household-characteristics.csv+1
computer_access	Indicator that the household has access to a computer / laptop	Binary (0/1)	1 if household has computer access, 0 otherwise	NSS Block 3 ICT variables; CMS ICT variables if presentR75252L02-Block-3-household-characteristics.csv+1
state_edu_expenditure_pc	Perâ€'capita public education expenditure for the householdâ€™s state and year	Numeric (continuous)	Indian Rupees per person per year	Derived by merging stateâ€'wise education expenditure from the Dataful public dataset with population denominatorsdataful
log_fee_burden_ratio	Natural logarithm of the fee burden ratio plus a	Numeric	Real number	Used to stabilise variance for regression models where appropriate

Mapping from original variables to derived analysis variables

This subsection explains how the analysis variables are constructed from the raw NSS and CMS fields and how household-level measures are derived from person-level records. It is important for transparency and reproducibility.

Household-level variables

Derived variable	Definition in analysis dataset	Original variable(s) and source
household_id	Concatenated unique key at household level used across rounds	CMS: scheduleid, surveyyear, nssregion, district, samplehhldno; NSS: survey ID and household number fields in Block 3CMSE80HH25.csv+1
survey_year	Reference survey year	CMS: surveyyear; NSS: survey round reference year
sector	Rural/urban classification	CMS: sector; NSS: sector variable in Block 3
nss_region	NSS region identifier	CMS: nssregion; NSS: region/stratum codes
district	District identifier	CMS: district; NSS: district code
household_size	Number of household members	CMS: householdsize; NSS: household size in Block 3
household_type	Type of household by main source of income	CMS: householdtype; NSS: household type code
religion	Religion of household head	CMS: religion; NSS: religion code
social_group	Social group / caste category	CMS: socialgroup; NSS: socialâ€'group code
any_member_attending_school	Whether any household member currently attends school	CMS: anymemberattendingschool; NSS: indicator from Block 5, aggregated to household level
num_members_attending_school	Count of schoolâ€'going members	CMS: nummembersattendingschool; NSS: count of persons with current attendance in Block 5
usual_monthly_consumption_expenditure	Usual monthly consumption on all items	CMS: usualmonthlyconsumptionexpenditure; NSS: usual monthly consumption expenditure variable in Block 3CMSE80HH25.csv+1
annual_household_consumption	Annual consumption expenditure used as denominator in burden ratio	Derived as 12 Ã— usual_monthly_consumption_expenditure for each household

Student- and education-expenditure variables (NSS 75th Round)

Derived variable	Definition in analysis dataset	Original variable(s) and source
course_fee_amount	Annual course/tuition fee per student	NSS Block 5: expenditure on course/tuition fees over the reference period, converted to annual amount where necessaryR75252L02-Block-3-household-characteristics.csv
books_stationery_amount	Annual expenditure on books and stationery	NSS Block 5: expenditure on books, stationery and related materials
transport_amount	Annual expenditure on transport for attending institution	NSS Block 5: transportâ€'related education expenditure
private_coaching_amount	Annual expenditure on private tuition / coaching	NSS Block 5: private tuition/coaching expenditure
other_education_expenditure	Other annual educationâ€'related expenditure per student	NSS Block 5: remaining education expenditure components (uniform, exam fees, etc.), aggregated
total_education_expenditure _student	Total annual education expenditure per student	Sum of all studentâ€'level education expenditure components listed above
school_type	Type of institution attended (public/private)	NSS Block 5 variable on management type of institution (e.g., government, governmentâ€'aided, private unaided)
education_level	Current level of education	NSS Block 5 variable on class/level of education

Household-level education-expenditure and burden variables

Derived variable	Definition in analysis dataset	Original variable(s) and source
total_education_expenditure_h h	Total annual education expenditure per household	For each household, sum total_education_expenditure_student across all student records in NSS Block 5 linked by household IDR75252L02-Block-3-household-characteristics.csv
fee_burden_ratio	Education fee burden ratio	For each household, compute total_education_expenditure_hh / annual_household_consumption
high_burden	High fee burden indicator used as the main classification target	Binary indicator, defined as 1 if fee_burden_ratio exceeds a chosen threshold (e.g., 0.10), 0 otherwise. The threshold will be justified with descriptive analysis and literature.
private_institution	Studentâ€'level indicator for private (vs public) institution	Derived from NSS school_type codes by mapping private unaided and private aided to 1, government institutions to 0
internet_access	Household ICT access indicator	NSS Block 3 CIP/ICT variables on internet access; CMS ICT variables if available. Converted to 1/0 after handling missing values.R75252L02-Block-3-household-characteristics.csv+1
computer_access	Household computer/laptop access indicator	NSS Block 3 variable on computer availability; CMS IT access fields if available. Converted to 1/0.
state_edu_expenditure_pc	Stateâ€'level public education expenditure per capita	Derived by merging stateâ€'wise and yearâ€'wise education expenditure from the Dataful dataset with state population denominators, then joining to each household by state and year.dataful

If additional derived variables such as log-transformed ratios, categorical bins (e.g., low/medium/high burden), or interaction terms (e.g., private_institution × rural) are used in the final models, they will be documented in the final report's extended data dictionary with explicit formulas.

GitHub repository and planned open materials

To comply with the QM 640 synopsis requirements regarding transparency and reproducibility, all permissible data and code will be shared through a public GitHub repository. The repository will not contain raw NSS microdata files, which are subject to usage conditions, but will instead include:

- **Data documentation and metadata**
    - A README.md at the root level explaining the project objectives, data sources (NSS 75th Round, CMS 80th Round, state education expenditure), and licensing/usage restrictions.
    - A data_dictionary.md file providing the full data dictionary for all variables used in the analysis, expanding on the summary given here.
- **Code for data preparation and modelling**
    - A code/ folder containing all Jupyter/Colab notebooks and Python scripts used for:
        - Importing and cleaning raw CMS and NSS files.
        - Constructing household- and student-level variables (including total_education_expenditure_hh, fee_burden_ratio, and high_burden).
        - Merging external public expenditure data.

        - Running descriptive statistics, visualisations, and model training/evaluation (logistic regression, tree-based models, explainability methods).
    - Notebooks will be structured to run end-to-end, with clear comments and section headings that mirror the methodology in the report.
- **Derived and example datasets**
    - A data_processed/ folder containing de-identified, analysis-ready derived datasets at household or aggregated levels (where permissible), for example:
        - hh_education_burden_derived.csv (with anonymised IDs and selected variables needed to reproduce the key figures and models).
    - No raw NSS microdata or personally identifiable information will be uploaded; instead, any user wishing to replicate the full pipeline will need authorised access to the original NSS files from the official data provider and can then run the shared scripts.
- **Results and documentation assets**
    - A results/ folder containing main tables and figures exported from the notebooks (e.g., model performance tables, feature importance plots, burden-ratio distributions).
    - A docs/ folder with PDF copies of the synopsis, interim report, and final report where allowed by course policy.
    - A screenshots/ subfolder including screenshots of the final folder and file structure, as explicitly required in the QM 640 synopsis template, to demonstrate proper organisation of the project materials.

A stable URL, https://github.com/sheeteshkumarrath/qm640-education-fee-burden-capstone has been created and all the required data and information would be added, and the same link will be included in the appendix of the final report and on the title slide of the capstone presentation.

Analytic Approach

The analytic approach will include the following major steps.

1. Data cleaning and descriptive analysis

- Import and clean the NSS Block 3, Block 5, Block 6 files and the CMS-E 2025 household file.
- Merge the relevant blocks and aggregate individual-level education expenditures to the household level.
- Handle missing values and outliers in key variables using clear rules (for example, dropping records with missing core variables and applying capping for extreme values where justified).
- Compute descriptive statistics (means, medians, distributions, cross-tabulations) for education expenditure, fee burden ratio, and household characteristics, stratified by rural or urban sector, social group, and institution type.

2. Feature engineering

- Construct the target binary variable high burden (1 if the education fee burden ratio exceeds the chosen threshold, 0 otherwise).

- Create predictive features such as log-transformed consumption expenditure, indicators for private institution, use of private coaching, number of school-going members, and distance-to-school categories.

3. Modelling

- Baseline statistical model – logistic regression to estimate the probability of high fee burden using the selected predictors.
- Machine learning models – tree-based models such as Random Forest and gradient-boosted trees (for example, XGBoost) to capture potential non-linear relationships and interactions.
- Evaluation – use accuracy, precision, recall, F1-score, and ROC-AUC for the classification task. For any continuous affordability-stress score (optional secondary analysis), use mean absolute error, root mean square error, and R-squared.

4. Explainability and interpretation

- Use model-based feature importance and model-agnostic methods, such as SHAP values or permutation importance, to identify which predictors contribute most to the high fee burden outcome.
- Present partial dependence plots or similar visualisations to show how the predicted risk of high fee burden changes with key variables such as consumption level, institution type, or private coaching.

5. Robustness and subgroup analysis

- Perform subgroup analysis across rural and urban sectors and major social groups where sample sizes permit.

- Conduct sensitivity checks for alternative definitions of "high fee burden" (for example, thresholds of 10 percent and 15 percent) to test the stability of the findings.

Solution to RQ1–RQ4 (Planned Methods)

RQ1: What proportion of households experience high education fee burden?

Techniques: Descriptive statistics and confidence-interval estimation for a single proportion. The education fee burden ratio will be computed at the household level, and households will be classified as "high burden" if the ratio exceeds the chosen threshold (for example, 0.10). Proportions and 95 percent confidence intervals will be reported overall and by key subgroups (rural or urban sector, social group, institution type).

Evaluation metrics: Point estimate of the proportion, confidence interval width, and comparisons across subgroups using chi-square tests where appropriate.

RQ2: How well do household characteristics predict high fee burden?

Techniques: Logistic regression as a baseline model to estimate the probability of high fee burden using predictors such as household size, household type, religion, social group, rural or urban sector, and usual household consumption expenditure. Regularised logit models may be considered if needed to handle multicollinearity or many predictors.

Evaluation metrics: Accuracy, precision, recall, F1-score, and ROC-AUC on a held-out test set, as well as confusion matrices to understand false positives and false negatives.

RQ3: How do education-related characteristics influence high fee burden?

Techniques: Extension of the logistic regression model from RQ2 by including education-related variables such as institution type, course level, medium of instruction, distance to school, and use of private coaching. Where relevant, chi-square tests or odds ratios will be

used to compare categories (for example, public vs. private institutions). Tree-based machine learning models (Random Forest, gradient-boosted trees) will also be explored to capture non-linear effects and interactions.

Evaluation metrics: Same classification metrics as RQ2, plus comparison of model performance before and after adding education-related features to see how much additional predictive power they provide.

RQ4: Which predictors are most important in explaining high education fee burden?

Techniques: Feature-importance analysis from tree-based models, along with model-agnostic explainability tools such as SHAP values or permutation importance, to identify which variables play the largest role in predicting high fee burden. Partial dependence or similar plots may be used to show how the predicted probability of high fee burden varies with key predictors (for example, consumption level, institution type, private coaching).

Evaluation metrics: Stability and consistency of feature-importance rankings across models, as well as qualitative interpretability for stakeholders. Classification metrics remain the same as for RQ2 and RQ3. In the full report, the evaluation metrics will be accompanied by their standard formulae and worked examples, as required by the course.

Recommendation and Applications

The primary target users of the findings from this capstone project are:

- Education policymakers at the state and central levels who design and monitor schemes such as scholarships, fee waivers, and other forms of financial support.

- Government departments and agencies working on school education and social welfare, who can use the model to identify segments of households that are most likely to experience high education fee burden.
- Non-governmental organizations and foundations focusing on education access and equity, who can use insights on key risk factors to design targeted interventions or awareness programmes.
- School managements and education planners, especially in low-fee private schools, who may want to understand how fee structures and associated costs impact household affordability.

By combining household and education factors linked to high fee burden, the study aims to support more evidence-based targeting of financial and non-financial interventions, so that limited resources can be directed towards households that are most likely to experience affordability stress.

References

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